

The Spaghetti Effect: Fracture Dynamics in Elastic Rods

Micah Day-O’Connell

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Abstract

We examine the dynamics of bent rods released at one end, inspired by Audoly and Neukirch’s 2005 work. Spaghetti rods are bent into semicircles, and released at one end. The propagation and reflection of flexural waves are tracked with a high speed camera and image analysis software. We obtained curvature values at each point on the rod for each frame of our video, highlighting the point of maximum curvature, and the “curvature center of mass”, a weighted average of curvature sites over the rod. Analyzing their dynamics with time, we found propagation speeds that matched the order of magnitude of theoretical expressions. We argue that center-of-mass tracking offers a more robust measure of flexural wave propagation in the presence of reflection and imprecise boundary conditions.

1 Introduction

Dry spaghetti, when bent at both ends, rarely breaks into two pieces. The proposed mechanism behind this effect involves an initial fracture, which releases a burst of flexural waves which increase the curvature along the rod, triggering secondary breakages. In their 2005 paper, Audoly and Neukirch tackled a seemingly unrelated problem: How does a rod’s curvature change when it is bent into a semicircle and released at one end? This simplified analysis replicates the spaghetti effect indirectly - the release is treated as the “initial break” which sends flexural waves traveling down the rod, some of which may result in a secondary break. Audoly and Neukirch’s bend-and-release model, with simple boundary conditions, yields a self-similar solution shown below:

$$\kappa(s, t) = 2\kappa_0 S\left(\frac{s}{\sqrt{2\pi\gamma t}}\right). \quad (1)$$

In this equation, κ is a curvature as a function of distance from released end of the rod, s , and time t . κ_0 is the initial value of curvature upon release, S is the Fresnel sine integral $S(x) = \int_0^x \sin(\frac{\pi}{2}y^2)dy$, and γ is a speed-like parameter

that governs dispersive propagation. We used Audoly and Neukirch’s calculated values of $\gamma = 0.521 \frac{m^2}{s}$ for Barilla No.1 and $\gamma = 0.521 \frac{m^2}{s}$ for Barilla No.5. [1]

Rather than describing a wave with constant velocity v where $s \propto vt$, but rather a dispersive wave where $s \propto \sqrt{\gamma t}$. We aimed to verify the dispersive nature of these flexural waves in a real experiment. Although our setup did not match the precise bend-and-release conditions of the theoretical analysis, much of our analysis involved reflections and no breakage, it offered an opportunity to observe dispersive waves in a more complex setting.

While answering a seemingly trivial question, an exploration of the spaghetti effect leads us into the realm of partial differential equations and dispersive spreading. PDEs with a 4-th derivative, (like the Kirchhoff beam equation which Audoly and Neukirch draw on) sometimes involve the characteristic $\sqrt{t}$ scaling for dispersive processes. Studying a simple system like bent pasta can lead us to understand the same mathematical structure that governs the physics of water waves or light pulses in optical fibers. Additionally, spaghetti illustrates a counterintuitive fact in elasticity - sometimes decreasing stress (by releasing a rod) can actually *increase* strain, the curvature that leads to breaks. A better understanding of this effect would be sought after by both materials scientists and engineers.

Section 2 discusses our experimental procedure and apparatus, as well as the image-analysis code involved in measuring curvature. Section 3 involves a derivation of the curvature propagation speeds from Audoly and Neukirch’s theoretical expression. Results are discussed in section 5.

2 Methods

As a partial investigation into the spaghetti effect, we tested Audoly and Neukirch’s theoretical model by attempting to replicate their setup and obtain curvature measurements along the rod.

2.1 Experimental Setup

We clamped an upright rod of spaghetti from the bottom, and placed it against a black background, as shown in Fig. 1. Wearing black gloves, we bent the rod into a semicircle and released. An Edgetronic SC2+ captured videos at a range of 2000-12000 FPS. At higher FPS values, a light was used to increase the visibility of our rod.

The black gloves and background made it so that the pixels on the spaghetti rod were the only light ones present in the video. While we aimed to replicate the bend-and-release scenario, we deviated from Audoly and Neukirch’s model by manually bending the rod by eye. As a result, we did not precisely replicate their initial conditions. Nonetheless, broad qualitative features of wave propagation, such as the $\sqrt{t}$ scaling discussed in section 5 were consistent with the dispersive behavior predicted by Audoly and Neukirch. Video stills illustrating the reflection are shown in Fig. 2.

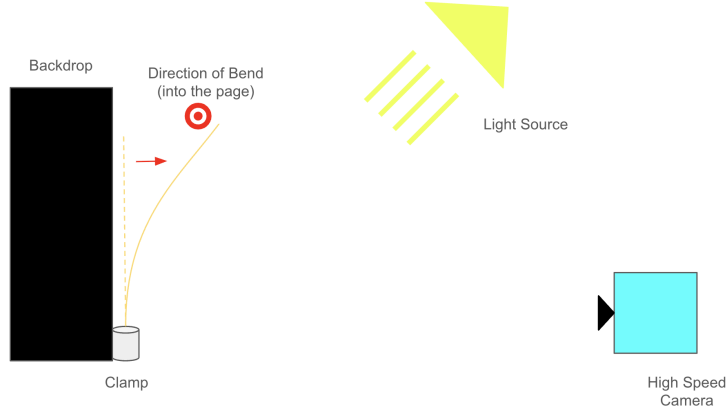


Figure 1: Our experimental setup. Rod was manually bent, which meant we deviated from Audoly and Neukirch’s initial conditions. High speed camera video was analyzed to obtain curvature data.

2.2 Data Collection and Curvature Calculation

Our data collection methods held two advantages over Audoly and Neukirch’s. Audoly and Neukirch’s method was to manually fit circles to their curved spaghetti to estimate curvature. Curvature, κ is defined as $\frac{1}{R}$ where R is the radius of an osculating circle, tangent to the curve, as shown in Fig.3 We used a more robust method of calculating curvature, discussed later, and our image analysis allowed us to calculate curvature continuously along the rod, rather than at a single point.

Videos were downloaded and passed to a series of Python scripts which estimated the position of our rod and calculated curvature. A repository of the code used can be found in the appendix. The first step in our video analysis was to “mask” each frame: pixels under a certain brightness threshold were set to black and the rest were set to white. Various OpenCV and scikit-image tools were used to remove any unwanted bright spots and create a “skeleton curve”: a one-pixel wide track through the center of the rod. Curvature can be calculated from the skeleton curve directly, but in order to smooth our values we fit a cubic spline to our skeleton curve, obtaining the parametric piecewise function $x(s)$ and $y(s)$ for our points as a function of arclength. To calculate curvature, we used the following expression:

$$\kappa = \frac{|x'y'' - y'x''|}{((x')^2 + (y')^2)^{3/2}}. \quad (2)$$

This version of curvature is equivalent to the osculating circle definition used by Audoly and Neukirch, albeit, significantly more robust. [2] Rather than taking the *signed* curvature we included an absolute value sign. whether the curvature was concave up or down was largely immaterial to our analysis - we

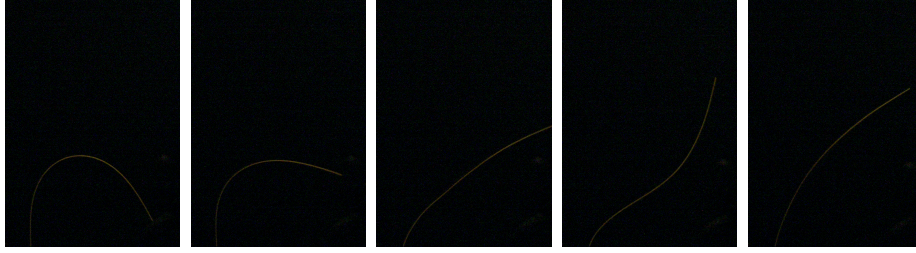


Figure 2: Still photos from Vid8.mov, linked in the appendix. Video was obtained at 5000 FPS, on a 24.5cm rod of Barilla No.5. **Left:** $t = 0s$. The rod is released after bending. **Center left:** $t = 3ms$. Curvature begins to propagate down the rod, the released end straightens. **Center:** $t = 6ms$. Before reflection, curvature has decreased significantly all across the rod. **Center right:** $t = 9ms$. Reflection results in two local peaks in curvature, one near the clamped end of the rod and one in the middle. **Right:** $t = 12ms$. Curvature wave reflects back to the free end, only one curvature peak remains.

were interested in the dispersive behavior of the entire wave’s energy. However, a signed curvature analysis could be useful for demonstrating other properties of the wave. More on this issue is discussed in section 5. Our data collection constituted a significant improvement on previous methods, as shown below in Fig. 4.

2.3 Outlier Detection

We obtained a large number of data points for each video frame. Due to differing light levels, reflective noise, or imperfections in our camera, certain individual frames were compromised. We aimed to tackle two common problems that were sources of outliers in our data. First, we sometimes encountered an artificially high value for curvature due to noise. Sometimes a small section of the clamp or a brief reflection from a glove or piece of dust was present in our frame, leading our code to mistakenly identify a “knee” in the curve and return extremely high curvature. Curvature energy should propagate slowly and continuously as the wave reflects, so we removed any sharp discontinuities in maximum curvature as outliers. The second problem was the opposite: instead of encountering bright spots, our camera gave noisy dark spot that inadvertently “split” the rod. Thus, any frames where a significantly smaller total arclength was detected indicated that our skeleton curve was only a partial capture. These frames were also marked as outliers. Even under ideal conditions, the best videos still had outliers, often with a frequency under five percent. In order to identify broader qualitative trends in our data, it was imperative that outliers were removed.

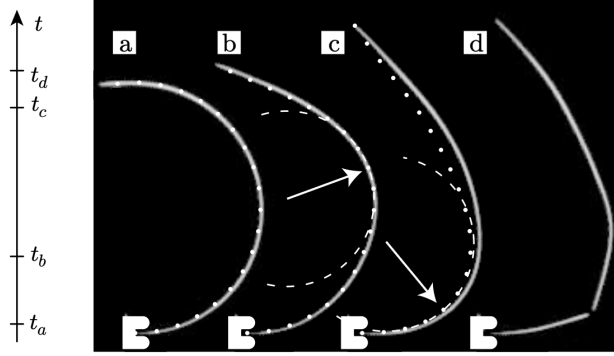


Figure 3: Figure 3 in Audoly and Neukirch’s paper. Dotted lines represent numerical simulations, dashed lines represent manual fitted circles, and arrows indicate the point of maximum curvature.

2.4 Image Analysis

Having obtained curvature data all along our rod, we saved various data points to a .csv file such as the point of maximum curvature, the value of curvature at that point, and the curvature center of mass, an “average” location of bending, defined as follows:

$$s_{cm}(t) = \frac{\sum s_i \kappa(s_i, t)}{\sum \kappa(s_i, t)}. \quad (3)$$

A center of mass analysis tracks how the bulk of the bending energy propagates over time. Since it incorporates the wavefront across the entire rod, it was less sensitive to noise and individual curvature maxima.

3 Theory

Equation 1 yields predictions for how the site of max curvature should propagate. Audoly and Neukirch obtained this expression by solving the Kirchhoff beam equation, a fourth-order partial differential equation that governs elastic rods. Their solution came in the form of a wave equation $u(\xi)$ where the characteristic variable $\xi = \frac{s}{\sqrt{\gamma t}}$ reflects the dispersive nature of flexural waves. The Fresnel sine integral, shown below in Fig. 5 arises in diffraction theory and serves as the basis for Audoly and Neukirch’s solution, Eq. 1.

The Fresnel sine integral peaks when $\xi = 2\sqrt{\pi}$ [1], or when $s = 2\sqrt{\pi\gamma t}$. This equation was the main source of theoretical predictions for wave propagation behavior. Using Audoly and Neukirch’s expressions for γ , we expected the point of maximum curvature to travel at a diffusive wave “speed” of $\frac{s}{\sqrt{t}} = 2.53$ for Barilla No.1 and $\frac{s}{\sqrt{t}} = 3.04$ for Barilla No.5. Importantly, these values should

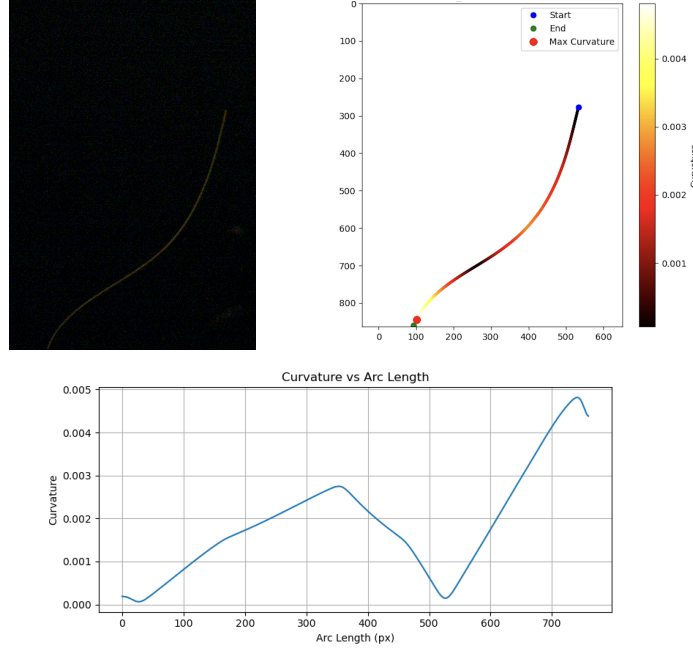


Figure 4: Data collection for frame 45 of Vid8.mov. **Top left:** The frame taken directly from the video. **Top right:** A curvature heatmap of that frame, after the skeleton and spline were computed. Curvature units, in pixels^{-1} , are largely immaterial. **Bottom:** A plot of curvature vs arclength. Again, units are immaterial, but similar plots were instrumental in computing the curvature center of mass and verifying the qualitative behavior of dispersive waves. Note that $s = 0$ is counted from the *free* end of the rod.

hold leading up to the first reflection, regardless of whether the spaghetti strand actually breaks. Since our rods were released with similar initial conditions to the theoretical prediction, we expected the site of max curvature to propagate linearly in $\sqrt{t}$, with a dispersive speed around 2.5 to $3 \frac{m}{\sqrt{s}}$.

When $\xi = 2\sqrt{\pi\gamma}$ the Fresnel sine function is maximized. This means that releasing the rod from a semicircle should increase its maximum curvature by a factor of $S(2\sqrt{\pi\gamma}) = 1.428$. As we will discuss later, the breaking behavior of the bend-and-release method was difficult to emulate - tracking curvature propagation and reflection *without* breakage was significantly more fruitful.

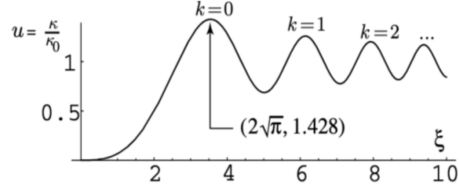


Figure 5: Figure 2b in Audoly and Neukirch’s paper, illustrating S , the Fresnel sine integral. This function governs the wave dynamics before reflection. The wave function $u(\xi) = 2S(\xi)$ is plotted on the vertical axis as dimensionless ratio of current to initial curvature, while $\xi = \frac{s}{\sqrt{\gamma t}}$ is plotted on the horizontal axis. Values for k are labels for maxima in the wavefront and are not wavenumbers.

4 Results

4.1 Breakage Effects

We were largely unable to replicate Audoly and Neukirch’s prediction for the curvature ratio between initial and maximum curvature before breakage. First, the bend-and-release setup was difficult to emulate, the rod often broke before it could be fully bent into a semicircle, or it did not break upon release. Furthermore, the clamp often provided a technical challenge: aggressive bend-and-release trials would often break the rod at clamp due to increased stress. We experimented with multiple barriers that would allow the rod to bend within the clamp such as clay and epoxy, but many of these did not lead to the desired effect. Many of the successful bend-and-release videos were taken under nonideal lighting conditions, and were difficult to analyze. Our single successful bend-and-release analysis of initial and maximum curvature is presented below.

For our single analysis we found an initial curvature of $0.00188(9)\text{pixels}^{-1}$ and a maximum curvature of $0.270(15)\text{pixels}^{-1}$. This estimated our curvature ratio at 1.4 ± 0.1 , containing Audoly and Neukirch’s value of 1.428 within our uncertainty.

4.2 Curvature Propagation Without Breakage

4.2.1 s_{max} - The Point of Maximum Curvature

Much more data was obtained in the case where the semicircle bend-and-release did *not* result in a breakage. We obtained curvature data up until the first reflection (where Eq. 1 holds) and many videos extended after multiple reflections. While Audoly and Neukirch predicted that s_{max} would propagate linearly with $\sqrt{t}$, this was difficult to obtain in practice. We often found that s_{max} followed more of a “step-function” behavior, and in linear regimes, values for s_{max} got caught on “ledges” as in Fig. 7.

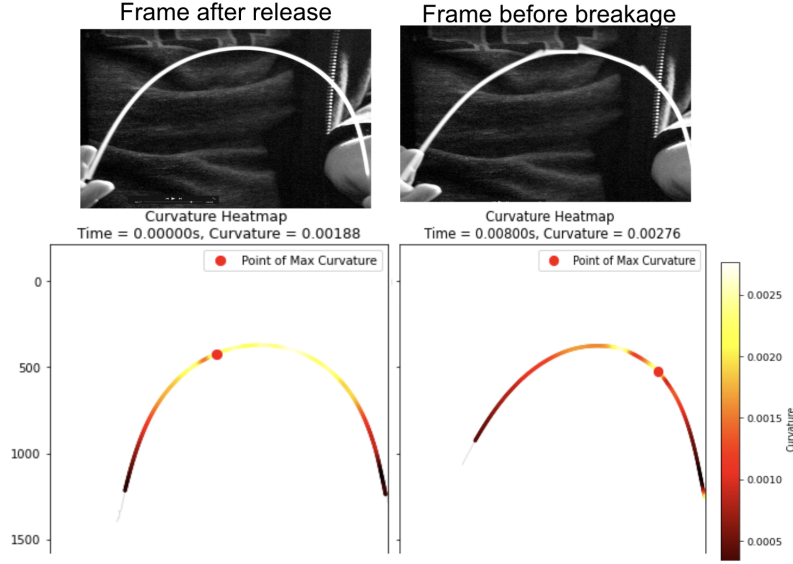


Figure 6: Two frames from a video demonstrating Audoly and Neukirch's prediction for $\frac{\kappa}{\kappa_0}$. No.5 spaghetti was used, video taken at 2000FPS. Note that these photos were taken under nonideal conditions, without lighting and black-out. Manual editing was used to mask unnecessary brightness or reflection that was misidentified as the rod. Curvature units are pixels^{-1} but are immaterial to an analysis of curvature *ratios*.

4.2.2 s_{cm} - The Curvature Center of Mass

Since the propagation of s_{max} was often obscured by the step-function behavior seen in Fig. 7, we turned to tracking the curvature center of mass, as defined in Eq.3 as a more reliable indicator of flexural waves. The center of mass followed a much more linear propagation with $\sqrt{t}$, as shown below in Fig.8.

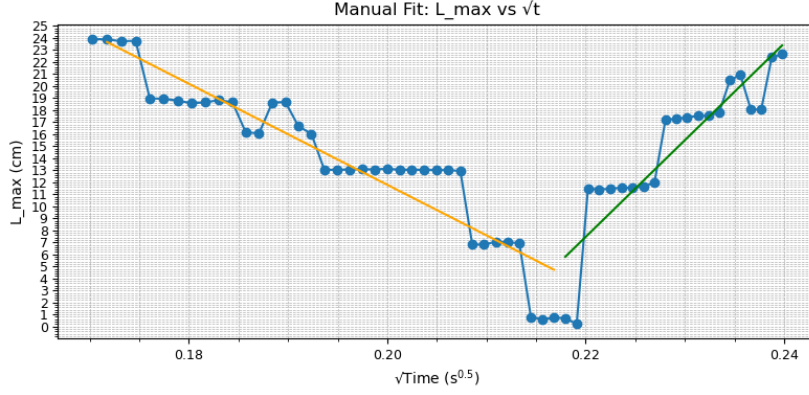


Figure 7: Plot of maximum curvature from a subsection of Vid8. Here L_{max} rather than s_{max} reflects the fact that units have been changed from arclength to length, making our dispersive wave speed determination meaningful. Linear segments are fit with slopes of $-4.21 \frac{m}{\sqrt{s}}$ ($R^2 = 0.89$) and $8\sqrt{s}$ ($R^2 = 0.86$) respectively. This “ledging” behavior was common for most bend-and-release videos.

Here, we see the characteristic behavior of s_{cm} for our non-breaking experiments. We often saw s_{cm} take a linear “random walk” with each new reflection. The slope values $\frac{s_{cm}}{\sqrt{t}}$ are presented in a table below:

Table 1: Slopes α and uncertainties for fits 1-7

Fit #	α	Slope (m/ $\sqrt{s}$)
1		-2.56 ± 0.13
2		6.6 ± 1.0
3		3.03 ± 0.81
4		1.6 ± 0.9
5		3.32 ± 0.24
6		-2.47 ± 0.07
7		5.64 ± 0.11

Table 2: Values for the linear fits in Fig. 8. Our slope $\frac{s_{cm}}{\sqrt{t}}$ took a “random walk” with each subsequent reflection, but remained on the expected order of magnitude.

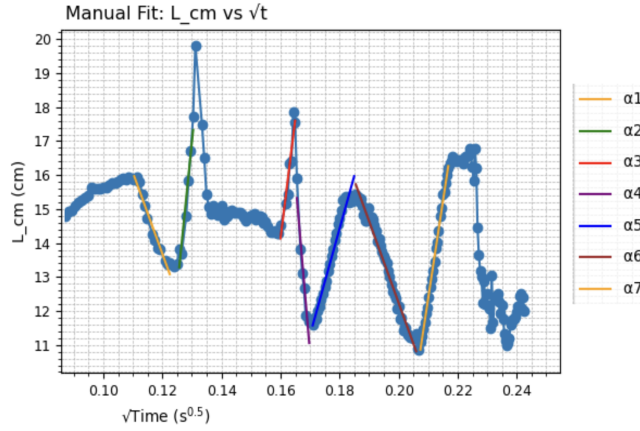


Figure 8: Curvature center of mass analysis for Vid24.mov, available in the appendix. Linear sections as s_{cm} propagates with $\sqrt{t}$ are fitted. A plot legend is provided on the side. Note that the marked sections are reflections, not covered by Audoly and Neukirch’s theory. Slopes with uncertainties for each linear fit segment are presented in table 2.

5 Discussion

Given the difficulties collecting data for the bend-and-release breakages, our value of 1.4 ± 0.1 for the curvature ratio is hardly conclusive, despite containing the expected theoretical value. Tracking the propagation of both s_{max} and s_{cm} was far more fruitful. Even though our experimental setup deviated from the ideal conditions studied by Audoly and Neukirch, key signatures of dispersive flexural wave propagation are still visible.

Tracking the point of maximum curvature gave rise to erratic “ledging” behavior, as seen in Fig. 7. Even when a linear trend was visible from the data, our max curvature sites were stubbornly stuck together. Several factors likely contributed to this. First, spots of brightness or noise on the rod itself can give rise to false maxima. It’s likely that any noise which evaded our outlier detection manifested as a “sticky” point of maximum curvature. Secondly, reflections of the flexural wave can lead to multiple peaks of curvature which “compete” with each other, leading to sudden jumps (see Fig. 4 for an example of multiple peaks after reflection). Third, the effects of our skeletonization and spline fitting are poorly understood, meaning that any uncertainty due to ledging may have been amplified.

Because of these issues, we turned to s_{cm} as a more reliable indicator of the wavefront’s propagation. As seen in Fig.3 and Table 2, we obtained reasonable estimates for center of mass propagation. Slope values for s_{cm} were on the same order of magnitude and the theoretical prediction for s_{max} . Vid24 was conducted with Barilla No.1, and fitted slopes were on the same order of

magnitude as the “breaking wave speed” of $2.53 \frac{m}{s^2}$. Since s_{cm} integrates curvature information from the entire rod, this analysis provides a better picture of the wave’s overall energy transport. This was another reason for looking at the absolute value of curvature rather than curvature itself. The wave energy care only about the magnitude of curvature, so we excluded the sign from out analysis. However, a signed curvature analysis could be useful for future work that looks at reflections, especially those that lead to multiple peaks with different curvature signs.

Uncertainty in curvature and the locations of max curvature was difficult to estimate. Our primary source of uncertainty was the camera’s resolution. However, any pixel error was later amplified by our skeleton curve, and the again by our spline fitting. We estimated a uncertainty of 5 percent: $\delta\kappa = 0.05\kappa$, but a more thoughtful treatment would have required a deeper understanding of spline fitting and more math than was useful for us to look into. Our uncertainties for the propagation of s_{max} and s_{cm} were obtained in two ways. In a preliminary analysis like Fig.8, curvature location was treated as exact and R^2 values quantified our credence in a linear relationship. Of course, s_{max} was already obscured by ledging, so we deemed reducing uncertainty to not be worthy of our time. For s_{cm} , we used a python package to find the lines of best fit. All slopes returned for the s_{cm} analysis in table 2 had R^2 values of 1.

The slope values in table 2 are either within the expected value, or higher. This seems counterintuitive, because our non-breaking waves should a *lower* magnitude of maximum curvature than breaking waves. Here, we can speculate a few reasons for this. First, when reflection occurs, some energy is lost due to vibration. While large curvature deviations require time to “build up” bending energy through constructive interference, smaller curvature deviations have a shorter “wait time” - they can propagate quickly before a large amount of energy is assembled. However, the theoretical behavior of flexural reflections are outside the scope of both this project and the 2005 article.

6 Conclusion

Using high speed video tracking and image analysis, we investigated the propagation of flexural waves in bent spaghetti, drawing on theoretical predictions from Audoly and Neukirch. While replicating the precise bend-and-release setup proved difficult, our single successful measurement supported the theoretical factor of 1.4, within experimental uncertainty.

More generally, tracking the propagation of the curvature center of mass allowed us to robustly measure propagation dynamics even in the presence of reflections, and imprecise initial conditions, both of which were not considered in Audoly and Neukirch’s ideal model. Qualitatively, we found that s_{cm} propagates linearly with $\sqrt{t}$, generally on the same order of magnitude as the predicted “speed” of $2.5 - 3 \frac{m}{\sqrt{s}}$.

One important lesson from this work is that simply obtaining more data does not always lead to a clearer physical understanding. On each frame, we collected

curvature data over roughly 600 pixels of arclength, with videos ranging from 100-1000 frames. While our curvature analysis revealed genuine trends, simply collecting orders of magnitude more data than in previous work did not always lead to better physical understanding. Future projects would have to take much more care with data collection and uncertainty estimation if this method is to be successful.

Nevertheless, the approximate linear propagation of s_{cm} with $\sqrt{t}$ and the fact that measured propagation speeds are generally on the same order of magnitude as the predicted $2.5-3 \frac{m}{\sqrt{s}}$ support the diffusive character of these flexural waves, a key part of Audoly and Neukirch's model.

7 Appendix

Code for the project can be found at this GitHub repository.
 Vid8.mov is in LabArchives and linked at this Google drive.
 Vid24.mov is in LabArchives and linked at this Google drive

8 Acknowledgments

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References

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- [2] Stewart, J. Multivariable calculus. (Cengage Learning, 2016).